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A deterministic verification strategy for electrostatic particle-in-cell algorithms in arbitrary spatial dimensions using the method of manufactured solutions ☆

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ABSTRACT

As simulations of kinetic plasmas continue to increase in scope and complexity, a rigorous and straightforward method for verifying particle-in-cell (PIC) implementations is necessary to ensure their correctness. In this paper, we present a deterministic method for the rigorous verification of multidimensional, multispecies, electrostatic particle-in-cell codes based on the method of manufactured solutions. Specifically, we prove that rigorous verification is possible through the exclusive examination of errors of grid quantities (i.e., moments and/or fields), allowing for a very light-weight and non-intrusive implementation in existing PIC codes. This is a marked improvement over earlier PIC verification approaches (only demonstrated with one species in 1D-1V), which rely on the comparison of cumulative distribution functions, and are invasive on the code base, introduce additional stochastic noise, are significantly more computationally expensive, and lack rigorous convergence properties. Interestingly, we show that different grid quantities feature different rates of convergence with the number of particles and mesh size, impacting the verification process. These theoretical results are confirmed numerically with a multi-species 2D-2V particle-in-cell code, including the ability of the method to detect order reduction due to an incorrect implementation.

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Particle-in-cell (PIC) methods are an important tool for the simulation of kinetic plasmas. These methods approximate a distribution function as a collection of particles, which are evolved according to equations of motion consistent with the characteristics of the Vlasov equation. While the fundamental PIC concept has a long history, recent years have seen tremendous increase in the complexity of these codes [1–7].

As these PIC algorithms are used in magnetic fusion, particle accelerator, and astrophysical applications, rigorously verifying the quality of these results is an increasingly critical issue. Classically, validation of these methods is done via code-to-code comparisons [8–10], the observation of specific physical features in more complex scenarios, or comparison against an analytical solution for extremely simple problems (e.g., Landau damping [11]). Clearly, none of these approaches is either comprehensive or rigorous enough. Recently, there has been some work on the application of the method of

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manufactured solutions (MMS) to the verification of electrostatic particle-in-cell schemes [12], based on the comparison of cumulative distribution functions (CDFs). However, this scheme suffers from various limitations, such as the considerable implementation and computation overhead to compute error norms in the CDF (which restricted the application of the method in Ref. [12] to 1D-1V), the noise introduced by a Monte Carlo treatment of said norms despite the use of quite large sample sizes ($\sim 10^6$), and the fact that measuring small errors in an integral (the CDF) does not guarantee small errors in the original function.

In this paper, we similarly employ the MMS to verify rigorously PIC implementations for the solution of the multidimensional electrostatic Vlasov-Poisson equation. In contrast to previous work, however, we show that only errors in grid quantities (moments and/or fields) need to be computed for verification. This seems intuitively true, since particles evolve along characteristics defined by these quantities. As a result, our verification approach saves orders of magnitude in computational complexity vs. existing alternatives [12], allowing in fact for a demonstration in multiple dimensions (2D-2V). Here, we provide a rigorous foundation on which to perform this verification, and present the interesting result that different grid quantities have different rates of convergence, which has direct implications for the appropriate choice of field for verification. Throughout, we restrict ourselves to the study of PIC schemes for the collisionless, electrostatic Vlasov-Poisson system. The spirit of the work presented here carries over to more complex physics scenarios - e.g. including electromagnetic effects - and the rigorous extension of our results will be the subject of future work.

The rest of the paper proceeds as follows. In Section 1, we briefly outline the particle-in-cell scheme we analyze throughout the manuscript. Section 2 covers several aspects of the method of manufactured solutions, including the construction of self-consistent solutions, how to test convergence, and issues in the computation of error norms for the distribution function used in earlier approaches. In section 3, we provide analysis that illustrates why errors in grid quantities are sufficient to verify a PIC implementation. Section 4 provides a cost comparison of the proposed scheme with previous efforts, demonstrating the significant efficiency advantage of our approach. In Section 5, numerical results demonstrate its effectiveness, including the ability to detect subtle coding errors. Finally, in section 6, we provide concluding remarks and outline future work.

1. Particle-in-cell methods

We focus on the PIC method for solving the multidimensional Vlasov-Poisson system, which in d-dimensions reads:

$$\frac{\partial f}{\partial t} + \mathbf{v}\cdot\nabla_x f + \frac{q}{m}\mathbf{E}\cdot\nabla_v f = 0, \tag{1a}$$

$$-\nabla^2\phi = \rho. \tag{1b}$$

Here, the independent variables $t \in \mathbb{R}$, $\mathbf{x} \in \mathbb{R}^d$, $\mathbf{v} \in \mathbb{R}^d$ are time, space, and velocity, respectively, and the dependent variables $f : \mathbb{R}\times\mathbb{R}^d\times\mathbb{R}^d \to \mathbb{R}$ and $\phi : \mathbb{R}\times\mathbb{R}^d \to \mathbb{R}$ are the species distribution function and electrostatic potential field, respectively, and m and q are mass and charge. Finally,

$$\rho = q\int_{\mathbb{R}^d} f\, d\mathbf{v}, \quad \text{and} \quad \mathbf{E} = -\nabla\phi$$

are the charge density and the electric field, respectively.

The particle-in-cell method approaches the numerical solution of (1) by first approximating the distribution function f as a number of discrete particles in phase space, and then modeling the time evolution of this distribution function through the movement of these particles according to forces determined by the electric field $\mathbf{E}$. Specifically, the algorithm proceeds as follows:

1. Sample $p^0(\mathbf{x},\mathbf{v})$ with N_p particles in $(\mathbf{x},\mathbf{v})$ phase-space (either randomly or with a quiet start technique). Assign to each particle its location in phase-space $(\mathbf{x}_p,\mathbf{v}_p)$ and a weight $w_p = f(\mathbf{x}_p,\mathbf{v}_p,0)/p^0(\mathbf{x}_p,\mathbf{v}_p)$ (in the particular case of $f(\mathbf{x}_p,\mathbf{v}_p,0) = p^0(\mathbf{x}_p,\mathbf{v}_p)$, $w_p = 1$).
2. Scatter the charge of each particle to grid locations to construct the charge density ρ_i at each grid node.
3. Solve the Poisson equation (1b) on this spatial grid to determine the electric field.
4. Gather the electric field at grid nodes $\mathbf{E}_i$ to each particle location to find $\mathbf{E}_p$.
5. Numerically propagate forward in time, from t to $t+\Delta t$, the location, velocity, and weight of each particle according to the equations of motion:

$$\frac{d\mathbf{x}_p}{dt} = \mathbf{v}_p, \quad \frac{d\mathbf{v}_p}{dt} = \frac{q}{m}\mathbf{E}_p, \quad \frac{dw_p}{dt} = 0. \tag{2}$$

6. Repeat steps 2 – 5 until $t = t_f$.

This approach has the benefit of mitigating the curse of dimensionality, requiring only a grid-based discretization of the d-dimensional spatial variable $\mathbf{x}$ and not the $2d$-dimensional phase space on which the distribution function f exists. These

savings are offset by the need to track the, potentially very many, particles in phase space, as well as the statistical noise associated with representing the distribution function as a collection of particles. The representation of the distribution function as a collection of particles has, in the past, been the primary obstacle to the application of rigorous verification methods. Later, we will show how it is possible to perform verification using only moments and/or fields.

2. Method of manufactured solutions

Verification using the method of manufactured solutions is based on the idea of solving numerically a new, contrived system that is related to the original system (1), and for which we know the exact solution. Verification is performed via the comparison of a theoretical convergence rate to the exact solution with an empirically measured convergence rate. If these convergence rates match, then we know both that the implemented numerical process is converging to the correct solution and that it matches the intended underlying algorithm.

The new system of equations to be solved is simply the previous Vlasov-Poisson equation (1) with the addition of a specific forcing term,

$$S_f = \frac{\partial f_M}{\partial t} + \mathbf{v}\cdot\nabla_x f_M + \frac{q}{m}\mathbf{E}_M\cdot\nabla_v f_M,$$
$$S_{\mathbf{E}} = \nabla\cdot\mathbf{E}_M - \rho_M.$$

Here f_M and $\mathbf{E}_M$ are the desired manufactured solution, and solve exactly the modified system:

$$\frac{\partial g}{\partial t} + \mathbf{v}\cdot\nabla_x g + \frac{q}{m}\mathbf{E}\cdot\nabla_v g = S_f, \tag{3a}$$
$$\nabla\cdot\mathbf{E} - \rho = S_{\mathbf{E}}. \tag{3b}$$

Unlike earlier studies [12], we insist here that $S_{\mathbf{E}} = 0$, since the errors hardest to identify occur in the particle-field coupling, and the artificial source terms may otherwise mask these errors. The addition of a source term in the Vlasov equation requires a small modification to the particle-in-cell procedure outlined in Section 1. Specifically, the equation of motion for the particle weight now has a nonzero right-hand side, becoming:

$$\frac{dw_p}{dt} = \frac{S_f(\mathbf{x}_p(t),\mathbf{v}_p(t),t)}{p^0(\mathbf{x}_p(0),\mathbf{v}_p(0))}. \tag{4}$$

2.1. Understanding theoretical convergence

The first step in the verification process is understanding the theoretical convergence rates of the quantities we may wish to measure numerically. Specifically, for particle-in-cell schemes for the Vlasov-Poisson system (1), those quantities are the distribution function f, potentially its corresponding cumulative distribution function F [12], the charge density ρ, the electric field $\mathbf{E}$, and the electrostatic potential ϕ.

Consider a hypothetical particle-in-cell implementation that employs a discrete Poisson solver and particle-to-grid and grid-to-particle interpolators of spatial order α, along with a particle pushing time integration scheme of temporal order β. We note for reference that most standard implementations employ second-order shape functions ($\alpha = 2$), and a second-order Boris particle push in time ($\beta = 2$).

The most direct and straightforward application of the MMS would be to measure errors in f. This is, after all, the fundamental unknown in the PDE system under study. As mentioned in the introduction, this is complicated by PIC's particle representation of the distribution function. There are, of course, numerous methods for reconstructing a probability density given some number of samples from that density [13]. Some of these methods lack clear convergence rate results, and all require prohibitively large particle numbers in the high-dimensional spaces (up to 6D) arising in PIC schemes. An illustrative example of such a method involves simply extending the configuration-space shape functions normally used in PIC to all of phase-space, allowing one to project particle properties onto a phase space grid with resolutions Δx and Δv in configuration and velocity space, respectively. The distribution function resulting from this procedure would have an error of the form

$$\varepsilon_f = C_1\Delta x^\alpha + C_2\Delta t^\beta + C_3\left(N_p\Delta x^d\Delta v^d\right)^{-1/2} + C_4\Delta v^\alpha + H.O.T., \tag{5}$$

where $H.O.T.$ is a shorthand for higher-order terms. Note that $(N_p\Delta x^d\Delta v^d)$ is proportional to the number of particles per *phase-space* cell, and the $\mathcal{O}(\Delta v^\alpha)$ term arises from the use of velocity-space shape-functions.

Performing a numerical convergence study on an error of this form is clearly intractable. Observing convergence in the spatial resolution requires small Δx, and an accurate distribution function reconstruction requires small Δv. These two combined force the use of an astronomically large number of particles to make the sampling error small. Indeed, directly studying convergence with this method would require a total particle number that is orders of magnitude larger than that required to run the underlying PIC scheme.

Table 1
Sampling error scaling of various field quantities with respect to the size of grid and number of particles in one-, two-, and three-physical dimensions.

Sampling Error Scalings	ρ	$\mathbf{E}$	ϕ
1-D	$(N_p\Delta x)^{-1/2}$	$N_p^{-1/2}$	$N_p^{-1/2}$
2-D	$\left(N_p\Delta x^2\right)^{-1/2}$	$\left(\lvert\log\Delta x\rvert/N_p\right)^{1/2}$	$N_p^{-1/2}$
3-D	$\left(N_p\Delta x^3\right)^{-1/2}$	$\left(N_p\Delta x\right)^{-1/2}$	$N_p^{-1/2}$

Thus, one is motivated to seek convergence metrics other than the direct use of f. One possible metric is the *cumulative* distribution function, denoted here by F, defined by one or more integrals of f over a portion of phase space. Being an integral quantity, F has a much more favorable error scaling:

$$\varepsilon_F = C_1\Delta x^\alpha + C_2\Delta t^\beta + C_3 N_p^{-1/2} + H.O.T.$$

This is the metric pursued in [12], but we argue in Section 3 that this approach – while clearly preferable to the naive strategy above – has noteworthy disadvantages.

We will propose the use of the grid quantities ρ, $\mathbf{E}$, and ϕ for verification instead. A necessary step in using these quantities as convergence metrics is to understand their theoretical convergence rates. Being a moment of the distribution function, it is straightforward to show that ρ has the error scaling: [14]

$$\varepsilon_\rho = C_1\Delta x^\alpha + C_2\Delta t^\beta + C_3\left(N_p\Delta x^d\right)^{-1/2} + H.O.T.$$

The scalings of errors in $\mathbf{E}$ and ϕ turn out to be more subtle. The sampling error arises from the randomness in the particle locations, which may be well-approximated by white noise. White noise is known to have a flat Fourier spectrum, but solving the Poisson equation to obtain ϕ dramatically reduces the amplitude of the large wave-number modes relative to ρ – formally, the ratio is $1/k^2$. Differentiating ϕ to obtain $\mathbf{E}$ increases the noise in each mode by a factor of k relative to ϕ, but this is still smaller than the noise in ρ.

This intuitive argument is formalized in Appendix A, in which we derive scalings for the sampling errors of ϕ and $\mathbf{E}$. The results of this analysis are summarized in Table 1, and will be confirmed by our numerical studies. Note that, as expected, $\mathbf{E}$ features smaller sampling errors than ρ, and ϕ has smaller errors than $\mathbf{E}$. It is perhaps surprising that this Fourier argument leads to the conclusion that the sampling errors *scale* differently, rather than just altering the constants in front of the scalings. Finally, we note that the derivation ignores the dynamics of the Vlasov system, focusing entirely on the Poisson equation. More study is needed to confirm the generality of these scalings, although our numerical results provide compelling evidence that they are at least valid in the manufactured-solution context.

2.2. Constructing a solution

Given our understanding of the theoretical convergence rates of the various particle-in-cell quantities, we now need to construct an empirical convergence rate for comparison. To do so, we require the ability to compute numerical errors, and so we seek to construct a collection $(f_M, \mathbf{E}_M, S_f, S_\mathbf{E})$ such that (3) is solved exactly. Moreover, this solution should be sufficiently smooth and non-singular, comprised of terms of similar magnitude, general enough to excite all of the terms present in (3), and specific enough to activate all special routines present in the PIC code being examined. These properties will ensure that observed numerical errors are not dominated by a single component of the particle-in-cell scheme or some pathological behavior of the chosen solution, and more importantly, self-consistent ($S_\mathbf{E} = 0$) solutions guarantee that every component of the particle-in-cell scheme is properly coupled and thoroughly exercised.

A straightforward way to construct such a collection is to begin by choosing a form for the distribution function. Specifically, we let $f(\mathbf{x},\mathbf{v},t) = f_x(\mathbf{x},t)f_v(\mathbf{v})$, and note that

$$\rho = q\int_{\mathbb{R}^d} f(\mathbf{x},\mathbf{v},t)d\mathbf{v} = qf_x(\mathbf{x},t)\int_{\mathbb{R}^d} f_v(\mathbf{v})d\mathbf{v}.$$

We then impose the additional constraint that

$$\int_{\mathbb{R}^d} f_v(\mathbf{v})d\mathbf{v} = 1,$$

and now given an electrostatic potential, ϕ, it is possible to construct a consistent f and $\mathbf{E}$ through the relations

$$f_x = -\frac{1}{q}\nabla^2\phi, \quad \text{and} \quad \mathbf{E} = -\nabla\phi.$$

While this process appears straightforward, we have ignored the important fact that f is a distribution function and should be non-negative everywhere. Choosing a potential which makes this true for a single species, and satisfies the relevant boundary conditions, can be challenging, especially in nontrivial problem geometries. For this reason, we consider multiple species. Specifically, we will make use of both negatively and positively charged particles and construct separate distribution functions for each. Let f^i and f^e represent the probability distribution of positively and negatively charged particles respectively so that we have the new system

$$\begin{aligned}\frac{\partial f^e}{\partial t} + \mathbf{v}\cdot\nabla_x f^e + \frac{q_e\mathbf{E}}{m_e}\cdot\nabla_v f^e &= S_{f^e},\\ \frac{\partial f^i}{\partial t} + \mathbf{v}\cdot\nabla_x f^i + \frac{q_i\mathbf{E}}{m_i}\cdot\nabla_v f^i &= S_{f^i},\\ \nabla\cdot\mathbf{E} &= \rho.\end{aligned}$$

It is now possible to construct a consistent set $(f^e, f^i, \mathbf{E}, S_{f^e}, S_{f^i})$ via the following process:

1. Choose ϕ such that the Poisson boundary conditions are satisfied.
2. Choose f_v^e and f_v^i such that $\int_{\mathbb{R}^d} f_v^e(\mathbf{v})d\mathbf{v} = 1$ and $\int_{\mathbb{R}^d} f_v^i(\mathbf{v})d\mathbf{v} = 1$.
3. Choose f_x^e and f_x^i such that $q_e f_x^e + q_i f_x^i = -\nabla^2\phi$ and $f_x^e, f_x^i \geq 0$ for all $\mathbf{x}$.
4. Compute $\mathbf{E} = -\nabla\phi$.

We will construct such a solution in Sec. 5.

Remark 1. While the modifications to the particle-in-cell procedure described in Section 1 required to account for time-varying particle weights and the existence of both positively and negatively charged particles appear dramatic, they are in fact relatively minor. The existence of both positive and negatively charged particles is an almost ubiquitious feature of particle-in-cell schemes, allowing our formulation to exercise this portion of a code. Similarly, time-varying weights are a commonly appearing feature of particle-in-cell implementations, and even where they do not already exist, modifications to account for it should be relatively straightforward.

2.3. Computing an empirical convergence rate

Armed with a suitable manufactured solution, it is possible to compute exact numerical errors, and with them an empirical convergence rate. We consider once again the hypothetical particle-in-cell implementation that employs a Poisson solver and particle-to-grid and grid-to-particle interpolators of spatial order α, along with a particle-pushing time-integration scheme of temporal order β.

Recall from section 2.1 that the theoretical error in a given quantity from the particle-in-cell scheme should have an expansion which looks like

$$\varepsilon = C_1\Delta x^\alpha + C_2\Delta t^\beta + C_3 N^{-1/2} + H.O.T, \tag{7}$$

where N is related to the number of particles (e.g., total number of particles, number of particles per grid cell, etc.). To simplify the convergence study, we follow the strategy given in [12] and define the refinement parameter h, a target order of accuracy p, and the reference parameters Δx_0, Δt_0 and N_0. If we now impose the relationship

$$h^p = \left(\frac{\Delta x}{\Delta x_0}\right)^\alpha = \left(\frac{\Delta t}{\Delta t_0}\right)^\beta = \left(\frac{N}{N_0}\right)^{-1/2}, \tag{8}$$

then the theoretical errors (7) can be rewritten as

$$\varepsilon = C_p h^p + \mathcal{O}(h^{p+1}).$$

In this way, an empirical convergence rate, $\hat{p}$, in h can be computed by simultaneously refining in space, time, and the number of particles according to (8). A comparison of this empirical order of convergence $\hat{p}$ to the theoretical order of convergence p will reveal the results of the verification.

3. Performing verification

Previous work on the application of the method of manufactured solutions to particle-in-cell schemes [12] focused on computing errors in the cumulative distribution function. Unfortunately, the nature of the particle-in-cell algorithm makes

this difficult, since the manufactured solution we know is an analytical distribution function, while the numerical solution we obtain is a collection of particles. Reference [12] solved this problem in the 1D-1V case by employing a generalization of the Kolmogorov-Smirnov statistic introduced by Peacock [15]. The underlying idea of the Kolmogorov-Smirnov statistic in one dimension is the construction of a so-called empirical distribution function (EDF)

$$G_N(x) = \frac{1}{N}\sum_{p=1}^{N}\theta(x - x_p), \quad \text{with} \quad \theta(x) = \begin{cases} 0 & \text{if } x < 0 \\ 1 & \text{otherwise} \end{cases}$$

from the collection of particles, and then comparing this empirical distribution function with the cumulative distribution function coming from the known analytic distribution function. The generalization of the empirical distribution function to higher dimensions is not difficult, though the comparison to the known cumulative distribution function becomes more complicated due to the fact that Kolmogorov-Smirnov statistic is not distribution-free [15]. The number of required comparisons is 2^n, where the mathematical dimension $n = 2d$ is twice the physical dimension of the problem.

For a three-dimensional problem, the mathematical dimension is six, and would require the comparison of 64 empirical and cumulative distribution functions. Each evaluation of the EDF requires a sum across all particles, and computing the norms of $\|F(\mathbf{x}, \mathbf{v}) - G_N(\mathbf{x}, \mathbf{v})\|$ exactly involves checking each "corner" – $(\mathbf{x}_i, \mathbf{v}_j)$ for each $i, j = 1, 2, \ldots, N_p$ – of the distribution function, a very expensive computation. Riva et al. [12] addressed this issue through the use of several clever Monte Carlo approximations to keep the cost of such comparisons reasonable. Unfortunately, such methods add more noise to the error measures, complicating the computation of convergence rates, and still require a very large number of Monte Carlo samples to control noise (Ref. [12] employed 10^6), making them expensive in practice. Moreover, a rigorous proof for the convergence rates of the Kolmogorov-Smirnov statistic generalized to higher dimensions does not exist, and their convergence has only been verified empirically throughout the literature.

Remark 2. We note that it is unclear if verifying errors in the cumulative distribution function is sufficient to ensure convergence in the distribution function. Consider the straightforward hypothetical example in which the error in the distribution function looks like

$$\delta f(x, v, t) = \cos(k \cdot x),$$

so that the error in its corresponding cumulative distribution function is given as

$$\delta F(x, v, t) = \frac{1}{k}\sin(k \cdot x).$$

Here, as k becomes large it is possible to measure very small errors in the cumulative distribution functions, while the errors in the actual distribution function remain $\mathcal{O}(1)$.

3.1. Errors in the electric field are sufficient

Since the primary issue in applying the method of manufactured solutions to particle-in-cell schemes is the difficulty presented by the distribution function and its particle-based approximation, here we seek to avoid the issue by computing errors only in grid quantities. Intuitively, since particle locations are evolved along characteristics wholly determined by the grid quantities ρ, $\mathbf{E}$, and ϕ, it seems reasonable that if errors in these quantities are approaching zero appropriately, then the error in our representation of the distribution function is similarly decreasing. The purpose of this section is to place this intuition on rigorous mathematical footing.

Our overarching strategy in developing this theory is to recognize that the particles in a PIC scheme necessarily represent a random sample from *some* probability density function. We can theoretically study this function's proximity to the solution of the Vlasov equation *without* explicitly constructing it in code. This approach is motivated by the fact that the goal of a PIC scheme is the computation of expected values over the Vlasov solution which serve to reduce its dimensionality – e.g. moments of f, averages over certain phase space coordinates – rather than f itself. These expectations may be computed accurately if the particles represent a sample of a distribution that closely approximates f, even without direct access to f itself.

We note that implicit in this approach is the use of a "noisy start" for the PIC scheme, in which initial particle states are sampled randomly. Users of a code may prefer some other initialization for their production runs, but this does not prevent use of a "noisy start" for verification purposes. Indeed, the PIC framework's components – particle push, grid-to-particle and particle-to-grid interpolation, and field solve – are all independent of the initialization procedure.

Also, consistent with our probabilistic interpretation, we will assume in this section, for simplicity, that the phase-space integrals of f and g are normalized to unity. Generalizing our arguments to remove this assumption is entirely trivial. The generalization only complicates notation, thereby muddying the presentation, so we avoid it here. We also treat the single-species case throughout, with generalization to multiple species also being trivial.

Our argument proceeds via a series of lemmas and theorems that establish that field quantities suffice for verification. The statements and proofs are simplified for the unforced Vlasov equation (1), so at each step we present our arguments in

this case first. These results may be of general interest in the theory of PIC convergence, but to apply directly to the MMS, we must consider the forced case (3) (with $S_{\mathbf{E}} = 0$). Each result is generalized to the forced case in a corollary.

We begin with the following lemma, which shows that errors in the continuum distribution function can be bounded by errors in the electric field.

Lemma 1 *(Continuum error bound). Consider two different distribution functions f and $\tilde{f}$ that are equal at $t = 0$ and satisfy (1), but are advected by different electric fields $\mathbf{E}$ and $\tilde{\mathbf{E}}$. Then the difference in these distribution functions at a point $T > 0$ in time can be bounded according to*

$$\|\delta f\|_2 (t = T) \leq T \left\{ \max_{\mathbf{x},\mathbf{v},t\leq T} |\nabla_v f| \right\} \left\{ \max_{t\leq T} \|\delta \mathbf{E}\|_2 \right\},$$

where $\delta f = f - \tilde{f}$, and similarly $\delta \mathbf{E} = \mathbf{E} - \tilde{\mathbf{E}}$.

Proof. The two distribution functions f and $\tilde{f}$ solve separately the two Vlasov equations,

$$\partial_t f + \mathbf{v} \cdot \nabla_x f + \mathbf{E} \cdot \nabla_v f = 0,$$

and

$$\partial_t \tilde{f} + \mathbf{v} \cdot \nabla_x \tilde{f} + \tilde{\mathbf{E}} \cdot \nabla_v \tilde{f} = 0.$$

Taking the difference of these two equations leads to

$$\partial_t \delta f + \mathbf{v} \cdot \nabla_x \delta f + \mathbf{E} \cdot \nabla_v \delta f + \delta \mathbf{E} \cdot \nabla_v f - \delta \mathbf{E} \cdot \nabla_v \delta f = 0.$$

We proceed by multiplying the whole equation by δf and integrating across the entirety of $\mathbf{x}$ and $\mathbf{v}$.

$$\frac{1}{2}\frac{d}{dt}\|\delta f\|_2^2 + \frac{1}{2}\int\limits_{\Omega\times\mathbb{R}^3} \nabla_x \cdot \left[\mathbf{v}(\delta f)^2\right] d\mathbf{x}d\mathbf{v} + \frac{1}{2}\int\limits_{\Omega\times\mathbb{R}^3} \nabla_v \cdot \left[\tilde{\mathbf{E}}(\delta f)^2\right] d\mathbf{x}d\mathbf{v} = -\int\limits_{\Omega\times\mathbb{R}^3} \delta f \delta \mathbf{E} \cdot \nabla_v f \, d\mathbf{x}d\mathbf{v}.$$

Here Ω is the spatial domain, and we have made use of the fact that $\tilde{\mathbf{E}} = \mathbf{E} - \delta\mathbf{E}$ and $\delta f \nabla \delta f = \nabla\left(\delta f^2\right)/2$.

The second integral on the left hand side is clearly zero by the divergence theorem in $\mathbf{v}$ and the assumption that both f and $\tilde{f}$ go to zero for large v. For reasonable boundary conditions (e.g., periodic or reflective), the first integral is also zero due to the divergence theorem in $\mathbf{x}$. Indeed, we have

$$\frac{1}{2}\int\limits_{\Omega\times\mathbb{R}^3} \nabla_x \cdot \left[\mathbf{v}(\delta f)^2\right] d\mathbf{x}d\mathbf{v} = \frac{1}{2}\int\limits_{\mathbb{R}^3} \mathbf{v} \cdot \left[\int\limits_{\partial\Omega} (\delta f)^2 \, d\mathbf{S}\right] d\mathbf{v}.$$

For periodic boundary conditions, this is clearly zero, for the inner integral is zero. For specular reflection, we have

$$\delta f(\mathbf{x}, \mathbf{v}, t) = \delta f(\mathbf{x}, \mathbf{v} - 2(\mathbf{v}\cdot\mathbf{n})\mathbf{n}, t)$$

for every $\mathbf{x}$ on the boundary, and the velocity-space integral can be shown to vanish. Therefore, with most of the left hand side vanishing, we have

$$\|\delta f\|_2 \frac{d}{dt}\|\delta f\|_2 = -\int\limits_{\Omega\times\mathbb{R}} \delta f \delta \mathbf{E} \cdot \nabla_v f \, d\mathbf{x}d\mathbf{v}.$$

Applying the Cauchy-Schwarz inequality to the right side and canceling terms, we have

$$\frac{d}{dt}\|\delta f\|_2 \leq \|\delta \mathbf{E} \cdot \nabla_v f\|_2 .$$

This, in turn, implies

$$\frac{d}{dt}\|\delta f\|_2 \leq \left\{ \max_{\mathbf{x},\mathbf{v}} |\nabla_v f| \right\} \|\delta \mathbf{E}\|_2$$

for appropriate definitions of the norms of vector-valued quantities. Thus, the integral over time can be bounded simply as:

$$\|\delta f\|_2 (t = T) \leq T \left\{ \max_{\mathbf{x},\mathbf{v},t\leq T} |\nabla_v f| \right\} \left\{ \max_{t\leq T} \|\delta \mathbf{E}\|_2 \right\}. \quad \square$$

Corollary 1 *(Continuum Error bound – forced case). Consider two different distribution functions g and $\tilde{g}$ that are equal at $t = 0$ and satisfy (3), but are advected by different electric fields $\mathbf{E}$ and $\tilde{\mathbf{E}}$. Then the difference in these distribution functions at a point $T > t_0$ in time can be bounded according to*

$$\|\delta g\|_2 (t = T) \leq T \left\{ \max_{\mathbf{x},\mathbf{v},t\leq T} |\nabla_v g| \right\} \left\{ \max_{t\leq T} \|\delta \mathbf{E}\|_2 \right\},$$

where $\delta g = g - \tilde{g}$, and similarly $\delta \mathbf{E} = \mathbf{E} - \tilde{\mathbf{E}}$.

Proof. The two distribution functions g and $\tilde{g}$ solve separately the two Vlasov equations,

$$\partial_t g + \mathbf{v} \cdot \nabla_x g + \mathbf{E} \cdot \nabla_v g = S,$$

and

$$\partial_t \tilde{g} + \mathbf{v} \cdot \nabla_x \tilde{g} + \tilde{\mathbf{E}} \cdot \nabla_v \tilde{g} = S.$$

Taking the difference of these two equations leads to

$$\partial_t \delta g + \mathbf{v} \cdot \nabla_x \delta g + \mathbf{E} \cdot \nabla_v \delta g + \delta \mathbf{E} \cdot \nabla_v g - \delta \mathbf{E} \cdot \nabla_v \delta g = 0.$$

The proof then proceeds exactly as that of Lemma 1. □

It bears noting that these results are only of use if $\nabla_v f$ (resp., $\nabla_v g$) is bounded for all time. The velocity-space filamentation arising in, for example, collisionless Landau damping may cause one to question whether this is really true in general. However, in the context MMS verification, we *choose* the exact solution. We are therefore free to choose a solution with bounded derivatives.

This lemma and corollary are the first steps to performing rigorous verification via field quantities, as they show that in the continuum case errors in f and g can be bounded by errors in $\mathbf{E}$. The next step is to understand what it means in this context to introduce a particle representation of the continuous distribution function. Specifically, we would like to know what relationship the particles at a future time have to the underlying distribution function they "approximate" at that same time. To do so we introduce the following notation:

Definition 1. By $\{\mathbf{x}_i, \mathbf{v}_i\} \sim f$ we will mean that the collection of particles $\{\mathbf{x}_i, \mathbf{v}_i\}$ are samples from the distribution function f. That is, for any subset of phase space A, we have

$$\mathbb{P}[(\mathbf{x}_i, \mathbf{v}_i) \in A] = \int_A f \, d\mathbf{x} d\mathbf{v}$$

for every i.

Additionally, if we further introduce a corresponding weight (or likelihood ratio)

$$w_i = \frac{g(\mathbf{x}_i, \mathbf{v}_i)}{f(\mathbf{x}_i, \mathbf{v}_i)},$$

we will say that

$$\{\mathbf{x}_i, \mathbf{v}_i; w_i\} \overset{\mathrm{f}}{\sim} g,$$

or in words that the particle collection $\{\mathbf{x}_i, \mathbf{v}_i; w_i\}$ is an importance sampling of g via f.

In light of this definition, we have the following lemma.

Lemma 2 *(Particles sample a continuum distribution). Let the vector field $\tilde{\mathbf{E}}(\mathbf{x}, t)$ and the initial distribution functions f^0 and p^0 be given. Let $\tilde{f}$ be the solution of the PDE*

$$\partial_t \tilde{f} + \mathbf{v} \cdot \nabla_x \tilde{f} + \tilde{\mathbf{E}} \cdot \nabla_v \tilde{f} = 0, \qquad \tilde{f}(t = 0) = f^0, \tag{9}$$

and $\tilde{p}$ be the solution of

$$\partial_t \tilde{p} + \mathbf{v} \cdot \nabla_x \tilde{p} + \tilde{\mathbf{E}} \cdot \nabla_v \tilde{p} = 0, \qquad \tilde{p}(t = 0) = p^0. \tag{10}$$

Further, let $\{\mathbf{x}_p(0), \mathbf{v}_p(0)\} \sim p^0$ be independent, identically distributed (iid) samples, and define $\{\mathbf{x}_p, \mathbf{v}_p\}$ at later times by

$$\begin{aligned}\frac{d\mathbf{x}_p}{dt} &= \mathbf{v}_p,\\ \frac{d\mathbf{v}_p}{dt} &= \tilde{\mathbf{E}}(\mathbf{x}_p, t).\end{aligned} \tag{11}$$

Then, at arbitrary times $T > 0$, we have

$$\left\{\mathbf{x}_p(T), \mathbf{v}_p(T)\right\} \sim \tilde{p}(T), \qquad w_p = \frac{f^0\left(\mathbf{x}_p(0), \mathbf{v}_p(0)\right)}{p^0\left(\mathbf{x}_p(0), \mathbf{v}_p(0)\right)}, \qquad \left\{\mathbf{x}_p(T), \mathbf{v}_p(T); w_p\right\} \overset{\tilde{p}}{\sim} \tilde{f}(T).$$

Proof. Define A_T as the time-advance map for the ODE system (11). That is,

$$(\mathbf{x}_p(T), \mathbf{v}_p(T)) = A_T(\mathbf{x}_p(0), \mathbf{v}_p(0)).$$

Note that this map is invertible by the time-reversibility of the ODE.

We begin the proof by showing that particles at later times are truly samples of the distribution function satisfying (10). To do so, we note that for an arbitrary phase-space subset B

$$\mathbb{P}\left[(\mathbf{x}_p(T), \mathbf{v}_p(T)) \in B\right] = \mathbb{P}\left[(\mathbf{x}_p(0), \mathbf{v}_p(0)) \in A_T^{-1}(B)\right] = \int\limits_{A_T^{-1}(B)} p^0\, d\mathbf{x}d\mathbf{v}. \tag{12}$$

The time-advance map A_T traces characteristics of the equation (9), and $\tilde{p}$ is constant along these characteristics by definition. That is to say, $\tilde{p}(\mathbf{x}, \mathbf{v}, T) = p^0\left(A_T^{-1}(\mathbf{x}, \mathbf{v})\right)$. Equivalently, $p^0(\mathbf{x}, \mathbf{v}) = \tilde{p}\left(A_T(\mathbf{x}, \mathbf{v}), T\right)$. Thus, making a change of variables in the integral appearing on the right of (12), we have

$$\int\limits_{A_T^{-1}(B)} p^0(\mathbf{x}, \mathbf{v})\, d\mathbf{x}d\mathbf{v} = \int\limits_B \tilde{p}(\mathbf{x}, \mathbf{v}, T)\, |\det A_T|\, d\mathbf{x}d\mathbf{v}. \tag{13}$$

The Liouville theorem, which states that phase-space volume is conserved, guarantees that $|\det A_T| = 1$. Thus, combining (12) and (13), we have

$$\mathbb{P}\left[(\mathbf{x}_p(T), \mathbf{v}_p(T)) \in B\right] = \int\limits_B \tilde{p}(T)\, d\mathbf{x}d\mathbf{v}.$$

Because this is true for arbitrary sets B, it implies that $\tilde{p}(T)$ is the probability density of $(\mathbf{x}_p(T), \mathbf{v}_p(T))$. That is,

$$\left\{\mathbf{x}_p(T), \mathbf{v}_p(T)\right\} \sim \tilde{p}(T).$$

We now rely on Definition 1 to define the particle weights

$$w_p = \frac{f^0(\mathbf{x}_p(0), \mathbf{v}_p(0))}{p^0(\mathbf{x}_p(0), \mathbf{v}_p(0))}$$

so that

$$\{\mathbf{x}_p(0), \mathbf{v}_p(0); w_p\} \overset{p^0}{\sim} f^0.$$

Moreover, the weights are unchanging in time, and since the particles track characteristics we have

$$w_p = \frac{f^0\left(\mathbf{x}_p(0), \mathbf{v}_p(0)\right)}{p^0\left(\mathbf{x}_p(0), \mathbf{v}_p(0)\right)} = \frac{\tilde{f}\left(\mathbf{x}_p(T), \mathbf{v}_p(T)\right)}{\tilde{p}\left(\mathbf{x}_p(T), \mathbf{v}_p(T)\right)}.$$

Thus, by Definition 1 we have that

$$\left\{\mathbf{x}_p(T), \mathbf{v}_p(T); w_p\right\} \overset{\tilde{p}}{\sim} \tilde{f}(T). \quad \square$$

We note that it is common-place to regard particles in PIC as "macro-particles", having a finite-width charge density given by S. The result above, in contrast, assumes the philosophical stance that particles are truly point particles. There is, in fact, no contradiction, as particles in PIC may just as well be understood as point particles, with S being an interpolation kernel associated with the grid rather than a physical charge density. This point of view is promoted in e.g. [14,16]. We are thus free in the result above to consider the particles in a PIC scheme as point particles sampled from a continuous distribution function.

While this result is interesting in the general PIC case, we are studying here the convergence of the modified system (3) which has a forcing term on the right-hand-side and a characteristic equation for a weight w_i. The following corollary generalizes this lemma to that case.

Corollary 2 *(Particles sample a continuum distribution – forced case). Let the vector field* $\tilde{\mathbf{E}}(\mathbf{x}, t)$ *and the initial distribution functions* g^0 *and* p^0 *be given. Let* $\tilde{g}$ *be the solution of the PDE*

$$\partial_t \tilde{g} + \mathbf{v} \cdot \nabla_x \tilde{g} + \tilde{\mathbf{E}} \cdot \nabla_v \tilde{g} = S, \qquad \tilde{g}(t=0) = g^0, \tag{14}$$

and $\tilde{p}$ *the solution of*

$$\partial_t \tilde{p} + \mathbf{v} \cdot \nabla_x \tilde{p} + \tilde{\mathbf{E}} \cdot \nabla_v \tilde{p} = 0, \qquad \tilde{p}(t=0) = p^0. \tag{15}$$

Further, let $\{\mathbf{x}_p(0), \mathbf{v}_p(0)\} \sim p^0$ *be iid samples, and define* $\{\mathbf{x}_p, \mathbf{v}_p\}$ *at later times by*

$$\begin{aligned} \frac{d\mathbf{x}_p}{dt} &= \mathbf{v}_p, \\ \frac{d\mathbf{v}_p}{dt} &= \tilde{\mathbf{E}}(\mathbf{x}_p, t). \end{aligned} \tag{16}$$

Additionally, consider a weight corresponding to each of the above particles that satisfies the following

$$\frac{dw_p}{dt} = \frac{S(\mathbf{x}_p(t), \mathbf{v}_p(t))}{p^0(\mathbf{x}_p(0), \mathbf{v}_p(0))}, \qquad w_p(0) = \frac{g^0(\mathbf{x}_p(0), \mathbf{v}_p(0))}{p^0(\mathbf{x}_p(0), \mathbf{v}_p(0))}. \tag{17}$$

Then, at arbitrary times $T > 0$*, we have*

$$\left\{\mathbf{x}_p(T), \mathbf{v}_p(T); w_p(T)\right\} \overset{\tilde{p}}{\sim} \tilde{g}(T).$$

Proof. We note that, by (15), $\tilde{p}$ is constant along the characteristics (16), which is to say that $d\tilde{p}(\mathbf{x}_p(t), \mathbf{v}_p(t), t)/dt = 0$, or $\tilde{p}(\mathbf{x}_p(t), \mathbf{v}_p(t), t) = p^0(\mathbf{x}_p(0), \mathbf{v}_p(0))$. Additionally, the method of characteristics applied to the forcing term of (14) leads to the characteristic equation

$$\frac{d\tilde{g}(\mathbf{x}_p(t), \mathbf{v}_p(t), t)}{dt} = S(\mathbf{x}_p(t), \mathbf{v}_p(t), t),$$

which by (17) implies:

$$\frac{d\tilde{g}(\mathbf{x}_p(t), \mathbf{v}_p(t), t)}{dt} = \frac{dw_p}{dt} p^0(\mathbf{x}_p(0), \mathbf{v}_p(0)) = \frac{dw_p}{dt} \tilde{p}(\mathbf{x}_p(t), \mathbf{v}_p(t), t).$$

Since $\tilde{p}$ is conserved along characteristics, it follows that

$$\frac{d\tilde{g}(\mathbf{x}_p(t), \mathbf{v}_p(t), t)}{dt} = \frac{d\left(w(\mathbf{x}_p(t), \mathbf{v}_p(t), t)\tilde{p}(\mathbf{x}_p(t), \mathbf{v}_p(t), t)\right)}{dt}.$$

Solving this ODE leads to

$$\tilde{g}(\mathbf{x}_p(T), \mathbf{v}_p(T), T) = w(\mathbf{x}_p(T), \mathbf{v}_p(T))\tilde{p}(\mathbf{x}_p(T), \mathbf{v}_p(T), T),$$

or

$$w(\mathbf{x}_p(T), \mathbf{v}_p(T)) = \frac{\tilde{g}(\mathbf{x}_p(T), \mathbf{v}_p(T), T)}{\tilde{p}(\mathbf{x}_p(T), \mathbf{v}_p(T), T)},$$

which, along with the fact that $\{\mathbf{x}_p, \mathbf{v}_p\} \sim \tilde{p}$ (i.e. Lemma 2) and Definition 1, implies that

$$\left\{\mathbf{x}_p(T), \mathbf{v}_p(T); w_p(T)\right\} \overset{\tilde{p}}{\sim} \tilde{g}(T). \quad \square$$

Armed with the knowledge that particles advected through time by an electric field represent a sample from the distribution function which solves the Vlasov equation (1a) containing that same electric field, it is possible to put everything together and show that verification can be performed through exclusive consideration of errors in the electric field.

Before doing so, however, there is a subtlety to be acknowledged. In the preceding lemmas and corollaries, the discussion has assumed an externally imposed electric field that is independent of the particles. This is what allows particle states to remain independent for all time. In an actual PIC simulation, of course, the electric field is determined self-consistently from the particles. As such, each particle has a small influence on the trajectory of other particles, and the trajectories are not strictly independent for any finite particle number. In related systems in which particles interact through a mean field (e.g., McKean-Vlasov equations), there are rigorous proofs that the trajectories tend toward independence as particle number diverges [17]. Extending this theory to a particle representation of the Vlasov-Poisson system is beyond the scope of this work, but we note that the analogies between the systems are exceedingly strong, providing compelling evidence that a similar result does in fact hold. As such, we will consider particle trajectories $(\mathbf{x}_i, \mathbf{v}_i)$ *approximately* independent for large particle number, and ignore the effects of non-independence in the subsequent results.

Theorem 1 *(Verification via the electric field). Consider a PIC simulation of the Vlasov-Poisson system (1), and denote the computed electric field by $\tilde{\mathbf{E}}$. Then, the particles are samples from a distribution function $\tilde{f}$, and we have*

$$\left\| f - \tilde{f} \right\|_2 = \mathcal{O}\left(\left\| \mathbf{E} - \tilde{\mathbf{E}} \right\|_2 \right). \tag{18}$$

That is, convergence in the distribution function may be no slower than the convergence in the electric field.

Proof. This is a straightforward consequence of Lemmas 1 and 2. We let $\tilde{\mathbf{E}}$ denote the electric field computed by the PIC scheme, extended to continuous time so that the numerical integration of the characteristic equations coincides with their analytic solution. Lemma 2 tells us that the distribution of which the particles are a sample – namely, $\tilde{f}$ – satisfies a Vlasov equation with field $\tilde{\mathbf{E}}$, and Lemma 1 implies that the difference between the analytic solution f and $\tilde{f}$ is bounded by the difference between the analytic $\mathbf{E}$ and $\tilde{\mathbf{E}}$. □

As before, this result may be interesting in the general context of PIC schemes, but is not directly applicable to the MMS due to the lack of a right-hand-side source term. The following corollary generalizes this result to the forced case.

Corollary 3 *(Verification via the electric field – forced case). Consider a PIC simulation of the forced Vlasov-Poisson system (3) (with exact solution g), and denote the computed electric field by $\tilde{\mathbf{E}}$. Then, the numerically integrated particles $\{\tilde{\mathbf{x}}_p, \tilde{\mathbf{v}}_p; \tilde{w}_p\}$ are an importance sample of an approximate distribution function $\tilde{h}$ via $\tilde{f}$ (the solution of (1a) employing the electric field $\tilde{\mathbf{E}}$), which converges to the exact solution g as:*

$$\left\| g - \tilde{h} \right\|_2 = \mathcal{O}\left(\left\| \mathbf{E} - \tilde{\mathbf{E}} \right\|_2 \right).$$

Proof. We first note that, for a given $\tilde{\mathbf{E}}$ and by Lemma 2, $\{\tilde{\mathbf{x}}_p(t), \tilde{\mathbf{v}}_p(t)\} \sim \tilde{f}$, and that for the given trajectories of the particles $\{\tilde{\mathbf{x}}_p(t), \tilde{\mathbf{v}}_p(t)\}$ there exists some $w(t)$ that satisfies the characteristic equation (4) exactly. If we define the distribution function $\tilde{h}$ corresponding to the numerical PIC solution for which

$$\{\tilde{\mathbf{x}}_p(t), \tilde{\mathbf{v}}_p(t); \tilde{w}_p(t)\} \overset{\tilde{\mathrm{f}}}{\sim} \tilde{h}.$$

Then by Corollary 2, we have:

$$\{\tilde{\mathbf{x}}_p(t), \tilde{\mathbf{v}}_p(t); w_p(t)\} \overset{\tilde{\mathrm{f}}}{\sim} \tilde{g},$$

and

$$w = \frac{\tilde{g}}{\tilde{f}}, \quad \text{and} \quad \tilde{w} = \frac{\tilde{h}}{\tilde{f}},$$

so that

$$w - \tilde{w} = \frac{\tilde{g} - \tilde{h}}{\tilde{f}}.$$

Together with the fact that our PIC method has temporal order β, this implies that

$$\|\tilde{g} - \tilde{h}\|_2 = \mathcal{O}(\Delta t^\beta).$$

Finally, according to Corollary 1 we have:

$$\|g - \tilde{g}\|_2 = \mathcal{O}(\|\mathbf{E} - \tilde{\mathbf{E}}\|_2).$$

Putting everything together we conclude that:

$$\|g - \tilde{h}\| \leq \|g - \tilde{g}\| + \|\tilde{g} - \tilde{h}\| = \mathcal{O}(\|\mathbf{E} - \tilde{\mathbf{E}}\|_2) + \mathcal{O}(\Delta t^\beta) = \mathcal{O}(\|\mathbf{E} - \tilde{\mathbf{E}}\|_2),$$

where the final equality is just a consequence of the fact that numerical errors in $\mathbf{E}$ themselves contain $\mathcal{O}(\Delta t^\beta)$ terms. This implies that the convergence of the approximate distribution function $\tilde{h}$ to the exact one, g, depends entirely on the convergence of $\mathbf{E}$, and so measuring convergence in the electric field guarantees the convergence of the distribution function. □

This result states exactly that if the theoretical convergence rate is observed in $\mathbf{E}$, then it is guaranteed in the distribution function as well! It thus suffices to use the electric field as the convergence metric when using the MMS.

3.2. Errors in the charge density are sufficient

Having shown that it is possible to perform verification using only the electric field, the question arises as to whether it is possible to perform verification using other grid quantities. Specifically, in two dimensions it is difficult to perform a convergence analysis on the electric field due to the logarithm appearing in its convergence rate (Table 1). As a first step toward this goal, we show that it is possible to bound errors in the electric field and electrostatic potential by errors in the charge density.

Lemma 3 *(Bounding errors in the potential via errors in the charge density). Consider the electrostatic potentials $\phi(\mathbf{x}, t)$, $\widetilde{\phi}(\mathbf{x}, t)$, and their difference $\delta\phi(\mathbf{x}, t)$, along with the charge densities $\rho(\mathbf{x}, t)$, $\widetilde{\rho}(\mathbf{x}, t)$, and their difference $\delta\rho(\mathbf{x}, t)$, with the corresponding potentials and charge densities related via the analytic Poisson equation in the case of ϕ*

$$\nabla^2\phi = -\rho \text{ on } \Omega, \text{ with } \phi = 0 \text{ on } \partial\Omega,$$

and a numerical approximation of the Poisson equation in the case of $\tilde{\phi}$

$$\nabla_h^2\tilde{\phi} = -\tilde{\rho}, \tag{19}$$

where ∇_h^2 is a numerical approximation of the Laplacian with order α with grid resolution h. That is, for arbitrary smooth functions u,

$$\nabla^2 u - \nabla_h^2 u = \mathcal{O}(h^\alpha). \tag{20}$$

Then, errors in the potential can be bounded via errors in the charge density as

$$\|\delta\phi\|_\infty \lesssim \|\delta\rho\|_\infty + \mathcal{O}(h^\alpha).$$

Proof. We first note that by the definition of ∇_h^2, we have

$$\nabla^2\widetilde{\phi} = -\widetilde{\rho} + r_\rho \text{ on } \Omega, \text{ with } \delta\phi = 0 \text{ on } \partial\Omega, \tag{21}$$

where $\|r_\rho\| \lesssim \mathcal{O}(h^\alpha)$.

We then note that due to the linearity of the operators, the potential and charge density differences satisfy the partial differential equation

$$\nabla^2\delta\phi = -\delta\widetilde{\rho} \text{ on } \Omega, \text{ with } \delta\phi = 0 \text{ on } \partial\Omega,$$

with $\delta\widetilde{\rho} = \rho - (\widetilde{\rho} + r_\rho)$. We then consider the function Ψ, the solution of

$$\nabla^2\Psi = -2 \text{ on } \Omega, \text{ with } \Psi = 0 \text{ on } \partial\Omega,$$

which for bounded domains Ω has a finite maximum that depends only on Ω, which we call $C_\Omega = \|\Psi\|_\infty$.

We now construct the super- and sub-solutions

$$\delta\phi_+ = \|\delta\widetilde{\rho}\|_\infty\Psi, \quad \delta\phi_- = -\|\delta\widetilde{\rho}\|_\infty\Psi.$$

It is easy to show that

$$\nabla^2(\delta\phi_+ - \delta\phi) = -2\|\delta\widetilde{\rho}\|_\infty - \delta\widetilde{\rho} \le 0,$$
$$(\delta\phi_+ - \delta\phi)|_{\partial\Omega} = 0.$$

The strictly nonpositive Laplacian implies via the maximum principle that $(\delta\phi_+ - \delta\phi)$ takes its minimum on the boundary. Since $(\delta\phi_+ - \delta\phi)$ vanishes on the boundary, it follows that $(\delta\phi_+ - \delta\phi) \ge 0$ everywhere. Putting everything together, we find:

$$\delta\phi \le \delta\phi_+ \le C_\Omega\|\delta\widetilde{\rho}\|_\infty.$$

If we follow through with the same logic with the sub-solution, $\delta\phi_-$ we get

$$\delta\phi \ge \delta\phi_- \ge -C_\Omega\|\delta\widetilde{\rho}\|_\infty.$$

These two facts taken together lead to the conclusion that

$$\|\delta\phi\|_\infty \le C_\Omega\|\delta\widetilde{\rho}\|_\infty \lesssim \|\delta\rho\|_\infty + \mathcal{O}(h^\alpha). \quad \square$$

Lemma 4 *(Bounding errors in the electric field via errors in the charge density). Consider the electrostatic potentials $\phi(\mathbf{x},t)$, $\widetilde{\phi}(\mathbf{x},t)$, and their difference $\delta\phi(\mathbf{x},t)$, the electric fields $\mathbf{E}(\mathbf{x},t)$, $\tilde{\mathbf{E}}(\mathbf{x},t)$ and their difference $\delta\mathbf{E}(\mathbf{x},t)$, along with the charge densities $\rho(\mathbf{x},t)$, $\widetilde{\rho}(\mathbf{x},t)$, and their difference $\delta\rho(\mathbf{x},t)$. The corresponding potentials, electric fields, and charge densities are related via the analytic Poisson equation in the case of ϕ*

$$\nabla^2\phi = -\rho \text{ on } \Omega, \text{ with } \phi = 0 \text{ on } \partial\Omega,$$

and a numerical approximation of the Poisson equation in the case of $\widetilde{\phi}$ (as in Lemma 3). Additionally, $-\nabla\phi = \mathbf{E}$ and $-\nabla_h\widetilde{\phi} = \tilde{\mathbf{E}}$, where ∇_h is a numerical approximation of the gradient with order α with grid resolution h. That is, for arbitrary smooth functions u,

$$\nabla u - \nabla_h u = \mathcal{O}(h^\alpha). \tag{22}$$

Then, the error in the electric field can be bounded via errors in the charge density as

$$\|\delta\mathbf{E}\|_2 \lesssim C\|\delta\rho\|_\infty + \mathcal{O}(h^\alpha).$$

Proof. We first note that by the definition of ∇_h, we have

$$\nabla\widetilde{\phi} = \tilde{\mathbf{E}} + r_{\mathbf{E}}, \tag{23}$$

where $\|r_{\mathbf{E}}\| \lesssim \mathcal{O}(h^\alpha)$. We then note that, due to the linearity of the operators, the differences in the field quantities also satisfy (as in Lemma 3)

$$\nabla^2\delta\phi = -\delta\widetilde{\rho} \text{ on } \Omega, \text{ with } \delta\phi = 0 \text{ on } \partial\Omega,$$

and $-\nabla\delta\phi = \delta\tilde{\mathbf{E}} = \mathbf{E} - \left(\tilde{\mathbf{E}} + r_{\mathbf{E}}\right)$. We now begin by multiplying both sides of the Poisson equation above by $\delta\phi$ and integrating over Ω, so that

$$\int_\Omega \delta\phi\nabla^2\delta\phi d\mathbf{x} = -\int_\Omega \delta\phi\delta\widetilde{\rho} d\mathbf{x},$$

$$-\int_\Omega |\nabla\delta\phi|^2 d\mathbf{x} + \int_{\partial\Omega} \delta\phi\nabla\delta\phi\cdot d\mathbf{S} = -\int_\Omega \delta\phi\delta\widetilde{\rho} d\mathbf{x},$$

$$\|\delta\tilde{\mathbf{E}}\|_2^2 \le \int_\Omega |\delta\widetilde{\rho}|\,|\delta\phi|\,d\mathbf{x} \le \|\delta\widetilde{\rho}\|_2\,\|\delta\phi\|_2.$$

Leveraging the fact that $\|\cdot\|_2 \le S\|\cdot\|_\infty$ (S is just the volume of the domain Ω) and employing Lemma 3 we conclude:

$$\|\delta\tilde{\mathbf{E}}\|_2 \le \sqrt{\|\delta\widetilde{\rho}\|_2\,\|\delta\phi\|_2} \le \sqrt{C_\rho\|\delta\widetilde{\rho}\|_\infty C_\phi\|\delta\phi\|_\infty} \le C\|\delta\widetilde{\rho}\|_\infty. \tag{24}$$

If we now consider in isolation the norm,

$$\|\delta\mathbf{E}\|_2^2 = \int |\delta\mathbf{E}|^2\,d\mathbf{x} = \int \left|\nabla\phi - \nabla_h\tilde{\phi}\right|^2 d\mathbf{x} = \int \left|\nabla\phi - \nabla\tilde{\phi} + \nabla\tilde{\phi} - \nabla_h\tilde{\phi}\right|^2 d\mathbf{x},$$

we can obtain, using the fact that $(a+b)^2 \le 2(a^2+b^2)$,

$$\|\delta\mathbf{E}\|_2^2 \le 2\int\left(\left|\nabla\phi - \nabla\tilde{\phi}\right|^2 + \left|\nabla\tilde{\phi} - \nabla_h\tilde{\phi}\right|^2\right)d\mathbf{x} \le 2\left\|\delta\tilde{\mathbf{E}}\right\|_2 + Ch^{2\alpha} \le \left(\sqrt{2}\left\|\delta\tilde{\mathbf{E}}\right\|_2^2 + \sqrt{C}h^\alpha\right)^2.$$

This fact together with equation (24) leads to

$$\|\delta\mathbf{E}\|_2 \lesssim \left\|\delta\tilde{\mathbf{E}}\right\|_2 + \mathcal{O}(h^\alpha) \lesssim \|\delta\widetilde{\rho}\|_\infty + \mathcal{O}(h^\alpha).$$

Finally, by noting that $\|\rho\|_\infty = \|\widetilde{\rho}\|_\infty + \mathcal{O}(h^\alpha)$ we conclude that

$$\|\delta\mathbf{E}\|_2 \lesssim C\|\delta\rho\|_\infty + \mathcal{O}(h^\alpha). \quad \square$$

We now have everything required to show that it is possible to perform verification using only the charge density, if desired.

Theorem 2 *(Verification via the charge density). Consider a PIC simulation of the Vlasov-Poisson system (1), and denote the computed electric field by $\tilde{\mathbf{E}}$. The particles are samples from a distribution function $\tilde{f}$, and we have*

$$\left\| f - \tilde{f} \right\|_2 = \mathcal{O}\left(\left\| \rho - \tilde{\rho} \right\|_\infty \right). \tag{25}$$

That is, convergence in the distribution function may be no slower than the convergence in the charge density.

Proof. Combining the results of Lemma 4 and Theorem 1 we have that

$$\left\| f - \tilde{f} \right\|_2 = \mathcal{O}\left(\left\| \mathbf{E} - \tilde{\mathbf{E}} \right\|_\infty \right) \lesssim \|\delta\rho\|_\infty + \mathcal{O}(h^\alpha).$$

Moreover, since the error expansion of $\delta\rho$ includes an $\mathcal{O}(h^\alpha)$ term we conclude that

$$\left\| f - \tilde{f} \right\|_2 = \mathcal{O}\left(\left\| \rho - \tilde{\rho} \right\|_\infty \right). \quad \square$$

Corollary 4 *(Verification via the charge density – forced case). Consider a PIC simulation of the forced Vlasov-Poisson system (3), and denote the computed electric field by $\tilde{\mathbf{E}}$. The particles $\{\tilde{\mathbf{x}}_p, \tilde{\mathbf{v}}_p; \tilde{w}_p\}$ are an importance sample of the distribution $\tilde{h}$ via $\tilde{f}$ (The solution of (1a) employing the electric field $\tilde{\mathbf{E}}$), and*

$$\left\| g - \tilde{h} \right\|_2 = \mathcal{O}\left(\left\| \rho - \tilde{\rho} \right\|_\infty \right). \tag{26}$$

Proof. Proof follows directly from Lemma 4 and Corollary 3 using the same logic as Theorem 2. $\square$

While an examination of Table 1 may lead to the desire to perform verification in the electrostatic potential, ϕ, due to its superior cost scaling (see next section), it is unfortunately not straightforward to show that this is reasonable. Specifically, consider a case similar to that in Remark 2. The same reasoning there casts doubt on the rigor of measuring errors in ϕ to bound errors in $\mathbf{E}$. For these reasons, we recommend performing verification via the electric field in one- and three-dimensions, and the charge density in two-dimensions (due to the awkwardness of the electric field's logarithmic scaling).

4. Cost comparison

We compare the computational complexity of our method, based on errors in grid quantities, and the method outlined in Section 3 and Ref. [12], based on errors in the cumulative distribution function. We note that the cost of these algorithms is determined by the cost to produce the PIC simulation, potentially many times, and the cost to compute the error norm. In addition to the differing costs of each simulation and error evaluation, there is the difference in scaling required for the convergence study in each case. For all methods, the cost of a PIC simulation scales like the number of particles N_p times the number of time-steps N_t.

We first consider the cost scaling of a method in d physical dimensions based on the computation of errors via the comparison of cumulative and empirical distribution functions [12]. Recall that a single evaluation of the empirical distribution function scales like the number of particles, and note that an exact evaluation of the norm $\|F^i(\mathbf{x},\mathbf{v}) - G^i_N(\mathbf{x},\mathbf{v})\|_\infty$ requires computing the difference of the distribution and empirical distribution functions at each "corner" $(\mathbf{x}_i, \mathbf{v}_j)$ for each $i, j = 1, 2, \ldots, N_p$, which scales like N_p^{2d}. In this way, we see that the cost of performing a PIC simulation and an exact error computation for the CDF method scales as:

$$C_F = \mathcal{O}\left(N_p \times N_t\right) + \mathcal{O}\left(N_p \times N_p^{2d}\right).$$

This scaling is dramatically improved by replacing the exact evaluation of $\|F^i(\mathbf{x},\mathbf{v}) - G^i_N(\mathbf{x},\mathbf{v})\|_\infty$ with a Monte Carlo approximation, at the cost of introducing more noise into the result. If a Monte Carlo approximation employing M samples is used, then the total cost of performing a PIC simulation and error computation scales as:

$$C_F^* = \mathcal{O}\left(R(M) \times N_p \times N_t\right) + \mathcal{O}\left(R(M) \times N_p \times M\right),$$

where $R(M)$ is the number of runs required to obtain a good result (i.e., decrease the Monte Carlo error below a given tolerance). According to (8):

$$h^p = \left(\frac{N_t}{N_{t,0}}\right)^{-\beta} = \left(\frac{N_p}{N_{p,0}}\right)^{-1/2}, \tag{27}$$

so that

$$N_p = \mathcal{O}\left(h^{-2p}\right), \quad N_t = \mathcal{O}\left(h^{-p/\beta}\right).$$

Introducing this scaling into the previous cost functions, we find

$$C_F = \mathcal{O}\left(h^{-2p-p/\beta}\right) + \mathcal{O}\left(h^{-4dp-2p}\right),$$

and

$$C_F^* = \mathcal{O}\left(R(M) \times h^{-2p-p/\beta}\right) + \mathcal{O}\left(R(M) \times h^{-2p} \times M\right).$$

In our method based on grid-quantity errors, however, the cost of an error evaluation is just a vector norm defined on the grid, which scales like the total number of mesh points, N_g. Therefore, the cost of performing a PIC simulation and error computation in the electrostatic potential ϕ goes like:

$$C_\phi = \mathcal{O}\left(N_p \times N_t\right) + \mathcal{O}\left(N_g\right).$$

From the sampling error scaling for ϕ in Table 1, we scale h with N_g as:

$$h^p = \left(\frac{N_g}{N_{g,0}}\right)^{-\alpha/d} \Rightarrow N_g = \mathcal{O}\left(h^{-pd/\alpha}\right),$$

with d the spatial dimensionality, leading to:

$$C_\phi = \mathcal{O}\left(h^{-2p-p/\beta}\right) + \mathcal{O}\left(h^{-pd/\alpha}\right).$$

The cost of performing a simulation and error evaluation using the electric field is identical to that of the electrostatic potential, though with a different scaling. In particular, we have:

$$C_{\mathbf{E}} = \mathcal{O}\left(N_p \times N_t\right) + \mathcal{O}\left(N_g\right), \tag{28}$$

but now (from Table 1, focusing on three dimensions for simplicity):

$$h^p = \left(\frac{1}{N_g^{1/d}} \frac{N_p}{N_{p,0}}\right)^{-1/2},$$

and therefore:

$$N_p = \mathcal{O}\left(h^{-2p} \times N_g^{1/d}\right) \sim \mathcal{O}\left(h^{-2p-p/\alpha}\right),$$

leading to

$$C_{\mathbf{E}} = \mathcal{O}\left(h^{-2p-p/\alpha-p/\beta}\right) + \mathcal{O}\left(h^{-pd/\alpha}\right).$$

Finally, the cost of performing a simulation and error evaluation using the charge density is

$$C_\rho = \mathcal{O}\left(N_p \times N_t\right) + \mathcal{O}\left(N_g\right),$$

with (from Table 1):

$$h^p = \left(\frac{1}{N_g} \frac{N_p}{N_{p,0}}\right)^{-1/2},$$

resulting in:

$$N_p = \mathcal{O}\left(h^{-2p} \times N_g\right) = \mathcal{O}\left(h^{-2p-pd/\alpha}\right),$$

and

$$C_\rho = \mathcal{O}\left(h^{-2p-pd/\alpha-p/\beta}\right) + \mathcal{O}\left(h^{-pd/\alpha}\right).$$

Assuming a method with $p = \alpha = \beta = 2$ in three-dimensions, we find:

$$C_\rho = \mathcal{O}\left(h^{-8}\right), \quad C_{\mathbf{E}} = \mathcal{O}\left(h^{-6}\right), \quad C_\phi = \mathcal{O}\left(h^{-5}\right),$$

and

$$C_F = \mathcal{O}\left(h^{-28}\right), \quad C_F^* = \mathcal{O}\left(R(M) \times h^{-5}\right) + \mathcal{O}\left(MR(M)h^{-4}\right).$$

The cost of grid-based verification methods is dominated by the cost of performing the PIC simulation, while the CDF-based verification approach is dominated by the cost of computing the errors. While the Monte Carlo approach appears to be the most competitive, it should be noted that $M \gg 1$ ($M = 10^6$ was used in [12]), and therefore it is quite expensive to use in practice. Moreover, we re-emphasize here that the cumulative distribution function used in [12] is an integrated quantity whose convergence may not be sufficient to guarantee proper convergence in f. If one is satisfied with an integrated quantity, it is simpler and more efficient to measure errors in ϕ instead. In contrast, the theorems proved above demonstrate that measuring convergence in $\mathbf{E}$ or ρ *is* sufficient to guarantee proper convergence in f, and in practice orders of magnitude cheaper than the CDF-based approach proposed in [12], enabling multidimensional verification, as we show in the next section.

5. Numerical results

We demonstrate the proposed verification scheme applied to a 2D-2V implementation of equation (1). We work in a periodic domain with a uniform grid, using leapfrog with bilinear shape functions and a spectral field solve. This sets $\alpha = \beta = 2$. The manufactured solution used throughout this section is:

$$
\begin{aligned}
F^e_M(\mathbf{x}, \mathbf{v}, t) &= f^0_v(u, v)\left(P \cdot S - P\sin(\pi t)\left[\frac{4\pi^2}{L_x^2}\sin\left(\frac{2\pi x}{L_x}\right) + \frac{4\pi^2}{L_y^2}\cos\left(\frac{2\pi y}{L_y}\right)\right]\right),\\
F^i_M(\mathbf{x}, \mathbf{v}, t) &= P \cdot S f^0_v(u, v)\\
\mathbf{E}_M(\mathbf{x}, t) &= \left\langle -\frac{2\pi}{L_x} P\sin(\pi t)\cos\left(\frac{2\pi x}{L_x}\right), \frac{2\pi}{L_y} P\sin(\pi t)\sin\left(\frac{2\pi y}{L_y}\right)\right\rangle,
\end{aligned}
$$

with

$$
f^0_v(u, v) = \frac{4}{\pi} u^2 v^2 e^{-u^2-v^2}, \quad S = \left(\frac{4\pi^2}{L_x^2} + \frac{4\pi^2}{L_y^2}\right), \quad \text{and} \quad P = \frac{1}{S L_x L_y}.
$$

Note that $f_x(\mathbf{x}, 0) = P \cdot S$. In these equations, $\mathbf{x} = (x, y) \in (0, L_x) \times (0, L_y)$ is the two-dimensional spatial domain, $\mathbf{v} = (u, v) \in (-\infty, \infty) \times (-\infty, \infty)$ is the two-dimensional velocity-space domain. In what follows, $L_x = L_y = 10$. For masses and charges, we consider $q_i = 1 = -q_e$ and $m_i/m_e = 1$. There results the following manufactured sources:

$$
\begin{aligned}
S_{f^e} = \frac{u v e^{-u^2-v^2}}{2000\pi^2}\Bigg(&2\sin(\pi t)\left(v\cos\left(\frac{\pi x}{5}\right)\left(\left(u^2-1\right)\sin(\pi t)\left(\sin\left(\frac{\pi x}{5}\right)+\cos\left(\frac{\pi y}{5}\right)\right)-2\left(1+2\pi^2\right)u^2+2\right)\right.\\
&+u\sin\left(\frac{\pi y}{5}\right)\left(-\left(v^2-1\right)\sin(\pi t)\left(\sin\left(\frac{\pi x}{5}\right)+\cos\left(\frac{\pi y}{5}\right)\right)+\left(2+4\pi^2\right)v^2-2\right)\Bigg)\\
&\left.-40\pi^2 u v\cos(\pi t)\left(\sin\left(\frac{\pi x}{5}\right)+\cos\left(\frac{\pi y}{5}\right)\right)\right),\\
S_{f^i} = \frac{u v\sin(\pi t)e^{-u^2-v^2}\left(\left(u^2-1\right)v\cos\left(\frac{\pi x}{5}\right)-u\left(v^2-1\right)\sin\left(\frac{\pi y}{5}\right)\right)}{500\pi^2}&.
\end{aligned}
$$

Remark 3. We note that while the specific manufactured solution given here is satisfactory for tests on our relatively generic PIC implementation, it may not be suitable in all cases. Specifically, if the PIC code being examined contains special features, such as specific routines for handling nonstandard geometries or complicating terms, it is important that the manufactured solution used exercises the code appropriately. We reiterate here the sentiments given in Section 2.2: a suitable solution should be sufficiently smooth and non-singular, comprised of terms of similar magnitude, general enough to excite all of the terms present in (3), and specific enough to activate all special routines present in the PIC code being examined.

In the convergence study below, the coarsest resolutions are $N_x = N_y = 12$, $\Delta t = 0.1$ and $N = 50$ particles per cell. The number of particles per cell, the size of the spatial discretization, and the time step are simultaneously refined according to the scaling laws in Table 1. Our implementation is second-order in time and space, and we have chosen the target order of accuracy, p, to also be second-order. Specifically, for ρ and ϕ we scale the numerical resolution parameters as:

$$
h^2 = \left(\frac{\Delta x}{\Delta x_0}\right)^2 = \left(\frac{\Delta y}{\Delta y_0}\right)^2 = \left(\frac{\Delta t}{\Delta t_0}\right)^2 = \left(\frac{N}{N_0}\right)^{-1/2}. \tag{29}
$$

For the electric field, assuming $\Delta x \sim \Delta y$, we use

$$
h^2 = \left(\frac{\Delta x}{\Delta x_0}\right)^2 = \left(\frac{\Delta y}{\Delta y_0}\right)^2 = \left(\frac{\Delta t}{\Delta t_0}\right)^2 = \left(\frac{N_p}{\log \Delta x}\cdot\frac{1}{N_{p,0}}\right)^{-1/2}. \tag{30}
$$

Particles are importance-sampled from $f(\mathbf{x}, \mathbf{v}, 0)$ using a convenient $p^0(\mathbf{x}, \mathbf{v})$ (here we choose p^0 to be uniform in $\mathbf{x}$ and Gaussian in $\mathbf{v}$) and are advanced in time via a modified leapfrog method

$$\mathbf{v}_p^{n+1/2} = \mathbf{v}_p^{n-1/2} + \Delta t \mathbf{E}_p^n, \tag{31a}$$

$$\mathbf{x}_p^{n+1} = \mathbf{x}_p^n + \Delta t\, \mathbf{v}_p^{n+1/2}, \tag{31b}$$

$$w_p^{n+1} = w_p^n + \Delta t \frac{S\left((\mathbf{x}_p^{n+1} + \mathbf{x}_p^n)/2, \mathbf{v}_p^{n+1/2}, t^{n+1/2}\right)}{p^0(\mathbf{x}_p(0), \mathbf{v}_p(0))}. \tag{31c}$$

Note that the use of time-centered evaluation of the right-side of the weight equation is important to maintain overall second-order accuracy of the scheme.

Fig. 1 shows the comparison of the empirical convergence rate of the charge density ρ, electric field $\mathbf{E}$, and the electrostatic potential ϕ. In all cases, we have performed simultaneous scaling according to the prescriptions above. Errors are computed as a simple L^2-norm over the field in the case of the scalar quantities ρ and ϕ,

$$\|\rho - \tilde{\rho}\|_2^2 = \frac{1}{N_x N_y} \sum_{i=1}^{N_x} \sum_{j=1}^{N_y} \left(\rho_{i,j} - \tilde{\rho}_{i,j}\right)^2, \quad \|\phi - \tilde{\phi}\|_2^2 = \frac{1}{N_x N_y} \sum_{i=1}^{N_x} \sum_{j=1}^{N_y} \left(\phi_{i,j} - \tilde{\phi}_{i,j}\right)^2,$$

and an L^2-norm of the magnitude in the case of the vector field $\mathbf{E}$,

$$\|\mathbf{E} - \tilde{\mathbf{E}}\|_2^2 = \frac{1}{N_x N_y} \sum_{i=1}^{N_x} \sum_{j=1}^{N_y} \left[\left([\mathbf{E}_x]_{i,j} - \left[\tilde{\mathbf{E}}_x\right]_{i,j} \right)^2 + \left(\left[\mathbf{E}_y\right]_{i,j} - \left[\tilde{\mathbf{E}}_y\right]_{i,j} \right)^2 \right]$$

These figures contain a line representing the empirical convergence of a properly implemented code, the empirical convergence of a code suffering from order reduction of the time integrator (we evaluate the position and velocity push at the same time, instead of using a properly staggered leapfrog integrator as in Eq. (31)), as well as a line representing the theoretical convergence behavior when applying the appropriate simultaneous scaling of Δx, Δy, Δt, and N. Fig. 1a shows the target order of accuracy for the correct implementation using ρ when scaling h according to Eq. (29) with the number of particles per cell, as predicted by the results in Table 1. Similarly, Fig. 1b shows the target order of accuracy for the correct implementation using $\mathbf{E}$ when scaling h according to Eq. (30), again in agreement with Table 1. Finally, Fig. 1c shows the target order of accuracy for the correct implementation using ϕ when scaling h according to Eq. (29) with the total number of particles, also in agreement with our analysis. All error measures show strong order reduction when employing the incorrect implementation. We remind the reader again that, while the convergence study in Fig. 1c is cheaper to run due to the more favorable scaling with h, it is not rigorously sufficient to guarantee convergence.

While the convergence study for the electric field in Fig. 1b is conclusive, we note that it was the most brittle measure of the three we considered owing to the presence of the logarithmic term in the convergence parameter scaling. It is our opinion that, due to these complications, the electric field should be avoided for the testing of convergence in two-dimensional PIC codes.

6. Conclusions

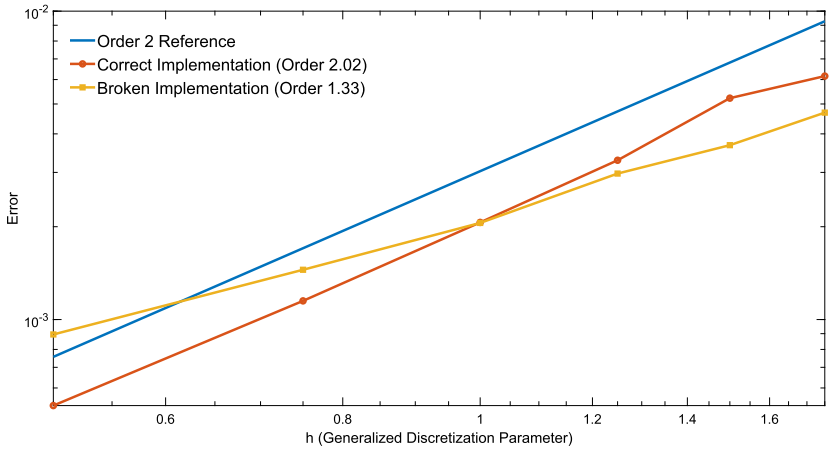
This work presents a verification strategy based on the method of manufactured solutions for electrostatic particle-in-cell schemes that only requires the computation of errors in grid quantities. In addition, we have presented some preliminary analysis exploring the differing convergence rates for different field quantities in an electrostatic particle-in-cell implementation. This analysis may not generalize to particle-in-cell solutions in general, but has been verified numerically in our numerical tests. The ability to perform verification with only grid quantities results in a dramatic efficiency improvement of the method of manufactured solutions vs. earlier studies [12]. In addition, the use of grid-based norms reduces the noise in the calculation of convergence rates, and avoids questions about the rigor of the statistical results needed for verification using the cumulative distribution function in higher dimensions. Future work will consider the extension of the method to electromagnetic PIC algorithms, as well as both electrostatic and electromagnetic gyrokinetic particle descriptions.

CRediT authorship contribution statement

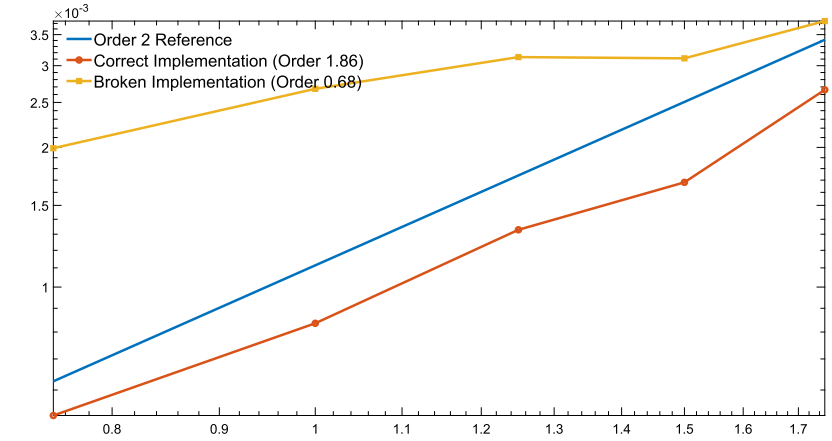
Paul Tranquilli: Formal analysis, Investigation, Software, Writing – original draft. **Lee Ricketson:** Formal analysis, Software, Writing – review & editing. **Luis Chacón:** Conceptualization, Supervision, Writing – review & editing.

Declaration of competing interest

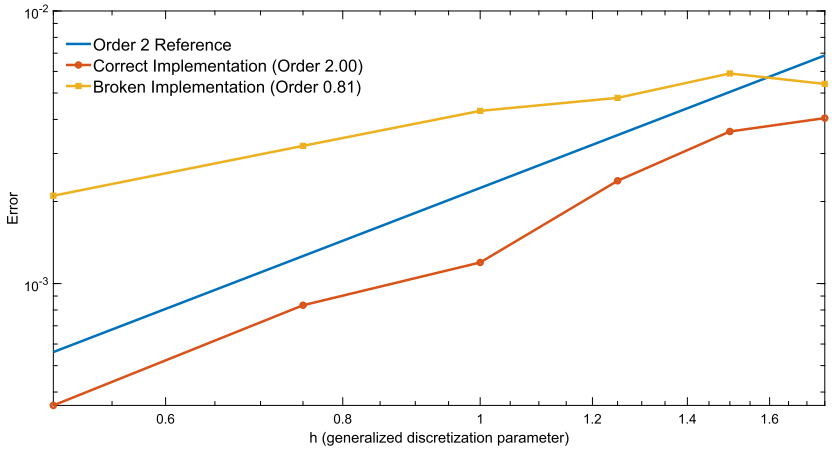
The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.



(a) Convergence study for ρ, scaling the number of particles per cell.



(b) Convergence study for $\mathbf{E}$, with logarithmic scaling as in Table 1.



(c) Convergence study for ϕ, scaling the total number of particles.

Fig. 1. Convergence studies for ρ, $\mathbf{E}$, and ϕ, employing the relevant scalings, for both a correct and broken implementation of the particle push time integration scheme.

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Appendix A. Deriving convergence rates of grid quantities

We derive the theoretical convergence rates of the grid quantities ρ, $\mathbf{E}$, ϕ, for a Vlasov-Poisson particle-in-cell code. These rates depend on the physical dimension of the problem, and so we derive each rate separately for one, two, and three dimensions. Derivations are performed for a spectral solver, but this is purely for simplicity - the same conclusions may be derived for, e.g., a second-order finite difference scheme. In addition, this analysis is static (there is no consideration for an evolution of time). We make no claims for the applicability of these results beyond the method of manufactured solutions.

A.1. One dimension

We consider specifically the case of periodic boundary conditions in a $[0, 2\pi]$ domain. Our grid has points $x_n = n\Delta x$, with $n = 0, \ldots, N_g - 1$ the cell index, with N_g the total number of cells, and $\Delta x = 2\pi/N_g$. We can compute the charge density from the particles at a grid point and decompose it into a deterministic and stochastic part

$$\rho_n = \bar{\rho}_n + s_n,$$

where $\bar{\rho}$ is noiseless (i.e. the limit as $N_p \to \infty$), and s is the contribution from particle noise. We begin with the Fourier transform of the charge density:

$$\widehat{\rho_k} = \sum_{n=0}^{N_g-1} \rho_n \exp\left\{-i\frac{2\pi n}{N_g}k\right\} = \sum_{n=0}^{N_g-1} \rho_n \exp\{-ikx_n\}.$$

The electrostatic potential in Fourier space from here is:

$$\widehat{\phi_k} = \frac{\widehat{\rho_k}}{\tilde{k}^2},$$

where $\tilde{k} = \min\{k, N_x - k\} \cdot \mathrm{sign}\{N_x/2 - k\}$ is a shorthand accounting for Fourier conventions (and similarly for $\tilde{\ell}$, $\tilde{j}$ in the multidimensional extensions below). We can then differentiate to get the electric field E:

$$\widehat{E_k} = \frac{-i\,\widehat{\rho_k}}{\tilde{k}}.$$

We know the stochastic portion of ρ is $s_n \sim \mathcal{N}(0, \sigma)$ with σ depending on the number of particles per cell N_p/N_g as:

$$\sigma \propto \sqrt{N_g/N_p},$$

so that the noise in the Fourier modes is $\widehat{s_k} \sim \mathcal{N}(0, \sigma\sqrt{N_g})$ – see e.g., [18]. Applying Plancherel's theorem we have

$$\sum_{m=0}^{N_g-1} \left|s_m^{E_x}\right|^2 = \frac{1}{N_g}\sum_{k=0}^{N_g-1} \left|\widehat{s}_k^{E_x}\right|^2 = \frac{1}{N_g}\sum_{k=1}^{N_g-1} \frac{|\widehat{s_k}|^2}{\tilde{k}^2}.$$

Taking expectations and noting that by symmetry the variances of $\mathbf{E}$ are equal at every grid point we have that

$$\left|\sigma_n^E\right|^2 = \frac{1}{N_g}\sum_{m=0}^{N_g-1} \left|s_m^{E_x}\right|^2 = \frac{\sigma^2}{N_g}\sum_{k=1}^{N_g-1} \frac{1}{\tilde{k}^2} \approx \frac{1}{N_p}\sum_{k=1}^{N_g/2} \frac{1}{k^2}.$$

We now consider the above as a Riemann sum for the integral

$$\int_1^{N_g/2} \frac{1}{r^2} dr = \left(1 - \frac{2}{N_g}\right) \approx 1.$$

So that

$$\sigma_E^2 \propto \frac{1}{N_p}.$$

Following the same process for ϕ we get that

$$\sum_{m=0}^{N_g-1}\left|s_m^{\phi}\right|^2=\frac{1}{N_g}\sum_{k=0}^{N_g-1}\left|\widehat{s}_k^{\phi}\right|^2=\frac{1}{N_g}\sum_{k=1}^{N_g-1}\frac{|\widehat{s}_k|^2}{\tilde{k}^4}.$$

Taking expectations and applying the symmetry argument

$$\left|\sigma_n^{\phi}\right|^2=\frac{\sigma^2}{N_g}\sum_{k=1}^{N_g-1}\frac{1}{\tilde{k}^4}\approx\frac{1}{N_p}\sum_{k=1}^{N_g/2}\frac{1}{k^4}.$$

Consider the above as a Riemann sum for the integral

$$\int_1^{N_g/2}\frac{1}{r^4}dr=\left(\frac{1}{3}-\frac{8}{N_g^3}\right)\approx\frac{1}{3}.$$

So that,

$$\sigma_{\phi}^2\propto\frac{1}{N_p}.$$

A.2. Two dimensions

In two-dimensions our grid requires two indices, and we similarly have two wave numbers in the Fourier transform. The 2-D discrete Fourier transform looks like:

$$\widehat{\rho}_{k\ell}=\sum_{m=0}^{N_x-1}\sum_{n=0}^{N_y-1}\rho_{mn}\exp\left\{-2\pi i\left(\frac{km}{N_x}+\frac{\ell n}{N_y}\right)\right\}.$$

Here, $N_g=N_xN_y$. We get the electrostatic potential in Fourier space from here via:

$$\widehat{\phi}_{k\ell}=\frac{\widehat{\rho}_{k\ell}}{\tilde{k}^2+\tilde{\ell}^2},$$

and then differentiate this to get **E**. So that

$$\widehat{E}_{x,k\ell}=\frac{-i\tilde{k}\widehat{\rho}_{k\ell}}{\tilde{k}^2+\tilde{\ell}^2},\qquad\widehat{E}_{y,k\ell}=\frac{-i\tilde{\ell}\widehat{\rho}_{k\ell}}{\tilde{k}^2+\tilde{\ell}^2}.$$

You can then just take the inverse transform to get **E**.

As before, the stochastic portion of ρ is $s_{mn}\sim\mathcal{N}(0,\sigma)$ with $\sigma\propto\sqrt{\frac{N_g}{N_p}}$, so that the noise in the Fourier modes is $\widehat{s}_{k\ell}\sim\mathcal{N}(0,\sigma\sqrt{N_g})$. Applying the Plancherel argument from above, with only minor modifications we have that

$$\sum_{m=0}^{N_x-1}\sum_{n=0}^{N_y-1}\left|s_{mn}^{E_x}\right|^2=\frac{1}{N_g}\sum_{k=0}^{N_x-1}\sum_{\ell=0}^{N_y-1}\left|\widehat{s}_{k\ell}^{E_x}\right|^2=\frac{1}{N_g}\sum_{k=1}^{N_x-1}\sum_{\ell=1}^{N_y-1}\frac{\tilde{k}^2\,|\widehat{s}_{k\ell}|^2}{\left(\tilde{k}^2+\tilde{\ell}^2\right)^2}.$$

As before, taking expectations and symmetry arguments combine to imply

$$\sigma_{E_x}^2=\frac{\sigma^2}{N_g}\sum_{k=1}^{N_x-1}\sum_{\ell=1}^{N_y-1}\frac{\tilde{k}^2}{\left(\tilde{k}^2+\tilde{\ell}^2\right)^2}\approx\frac{1}{N_p}\sum_{k=1}^{N_x/2}\sum_{\ell=1}^{N_y/2}\frac{k^2}{\left(k^2+\ell^2\right)^2}.\tag{A.1}$$

Call

$$\mathcal{S}=\sum_{k=1}^{N_x/2}\sum_{\ell=1}^{N_y/2}\frac{k^2}{\left(k^2+\ell^2\right)^2}=\sum_{k=1}^{N_x/2}\sum_{\ell=1}^{N_y/2}\frac{\ell^2}{\left(k^2+\ell^2\right)^2},$$

with the two expressions being equal by symmetry. We have

$$\mathcal{S}=\sum_{k=1}^{N_x/2}\sum_{\ell=1}^{N_y/2}\frac{k^2+\ell^2-\ell^2}{\left(k^2+\ell^2\right)^2}=\sum_{k=1}^{N_x/2}\sum_{\ell=1}^{N_y/2}\frac{1}{k^2+\ell^2}-\mathcal{S}.$$

Thus,

$$\mathcal{S} = \frac{1}{2}\sum_{k=1}^{N_x/2}\sum_{\ell=1}^{N_y/2}\frac{1}{k^2+\ell^2}.$$

We can consider $\mathcal{S}$ as a Riemann sum for the integral

$$\frac{1}{2}\int_1^{N_x/2}\int_1^{N_y/2}\frac{1}{x^2+y^2}\,dxdy,$$

which is itself comparable to the integral (for simplicity, let's assume $N_x = N_y = \sqrt{N_g}$):

$$\frac{1}{2}\int_1^{\sqrt{N_g}/2}\int_0^{2\pi}\frac{1}{r^2}r\,d\theta dr = \pi\int_1^{\sqrt{N_g}/2}\frac{1}{r}\,dr = \pi\log\frac{\sqrt{N_g}}{2}.$$

Finally, substituting this scaling into (A.1) gives

$$\sigma_{\mathbf{E}}^2 \propto \frac{\log N_g}{N_p}.$$

Following the same process for ϕ we get

$$\sum_{m=0}^{N_x-1}\sum_{n=0}^{N_y-1}\left|s_{mn}^{\phi}\right|^2 = \frac{1}{N_g}\sum_{k=0}^{N_x-1}\sum_{\ell=0}^{N_y-1}\left|\widehat{s}_{k\ell}^{\phi}\right|^2 = \frac{1}{N_g}\sum_{k=1}^{N_x-1}\sum_{\ell=1}^{N_y-1}\frac{|\widehat{s}_{k\ell}|^2}{\left(\tilde{k}^2+\tilde{\ell}^2\right)^2}.$$

Taking the expectation we get

$$\sigma_{\phi}^2 = \frac{1}{N_g}\sum_{m=0}^{N_x-1}\sum_{n=0}^{N_y-1}\left|s_{mn}^{\phi}\right|^2 = \frac{\sigma^2}{N_g}\sum_{k=1}^{N_x-1}\sum_{\ell=1}^{N_y-1}\frac{1}{\left(\tilde{k}^2+\tilde{\ell}^2\right)^2} \sim \frac{1}{N_p}\sum_{k=1}^{N_x-1}\sum_{\ell=1}^{N_y-1}\frac{1}{\left(\tilde{k}^2+\tilde{\ell}^2\right)^2}.$$

Considering the above as a Riemann integral

$$\int_1^{N_x/2}\int_1^{N_y/2}\frac{1}{(x^2+y^2)^2}\,dxdy,$$

which is comparable to

$$\int_1^{\sqrt{N_g}/2}\int_0^{2\pi}\frac{1}{r^4}r\,d\theta dr = \pi\int_1^{\sqrt{N_g}/2}\frac{1}{r^3}\,dr = \pi\left(\frac{1}{2}-\frac{2}{N_g}\right)\approx\frac{\pi}{2}.$$

We conclude:

$$\sigma_{\phi}^2 \propto \frac{1}{N_p}.$$

A.3. Three-dimensions

As above, we have that

$$\widehat{\phi}_{k\ell} = \frac{\widehat{\rho}_{k\ell}}{\tilde{j}^2+\tilde{k}^2+\tilde{\ell}^2},$$

and then differentiate this to get **E**. So that

$$\widehat{E}_{x,jk\ell} = \frac{-i\tilde{j}\widehat{\rho}_{jk\ell}}{\tilde{j}^2+\tilde{k}^2+\tilde{\ell}^2}, \qquad \widehat{E}_{y,jk\ell} = \frac{-i\tilde{k}\widehat{\rho}_{jk\ell}}{\tilde{k}^2+\tilde{\ell}^2}, \qquad \widehat{E}_{z,jk\ell} = \frac{-i\tilde{\ell}\widehat{\rho}_{k\ell}}{\tilde{j}^2+\tilde{k}^2+\tilde{\ell}^2}.$$

Then following the same logic as for two dimensions: Plancherel, taking expectations, exploiting symmetry, and noting that ρ is $s_{mnq} \sim \mathcal{N}(0, \sigma)$ with $\sigma \propto \sqrt{N_g/N_p}$, so that the noise in the Fourier modes is $\widehat{s}_{jk\ell} \sim \mathcal{N}(0, \sigma\sqrt{N_g})$, with $N_g = N_x N_y N_z$, we obtain

$$\sigma_{E_x}^2 \approx \frac{1}{N_p} \sum_{j=1}^{N_x/2} \sum_{k=1}^{N_y/2} \sum_{\ell=1}^{N_z/2} \frac{j^2}{\left(j^2+k^2+\ell^2\right)^2}.$$

Calling the triple sum $\mathcal{S}_3$ and employing the same symmetry-type argument as for the two-dimensional case gives

$$\mathcal{S}_3 = \frac{1}{3} \sum_{j=1}^{N_x/2} \sum_{k=1}^{N_y/2} \sum_{\ell=1}^{N_z/2} \frac{1}{j^2+k^2+\ell^2}.$$

Considering $\mathcal{S}_3$ as a Riemann sum for the integral

$$S_3 \approx \frac{1}{3} \int_1^{N_x/2} \int_1^{N_y/2} \int_1^{N_z/2} \frac{1}{x^2+y^2+z^2}\, dxdydz,$$

which is comparable to

$$S_3 \sim \int_1^{\sqrt[3]{N_g}/2} \int_0^{2\pi} \int_0^{\pi} \frac{1}{r^2} r^2 \sin\phi\, d\phi d\theta dr = 2\pi \int_1^{\sqrt[3]{N_g}/2} dr \sim \sqrt[3]{N_g}.$$

Putting everything together, we see that

$$\sigma_{\mathbf{E}}^2 \propto \frac{\sqrt[3]{N_g}}{N_p} = \frac{N_x}{N_p}.$$

For ϕ we have, after applying Plancherel, taking expectations, and exploiting symmetry, that

$$\sigma_\phi^2 \approx \frac{1}{N_p} \sum_{j=1}^{N_x/2} \sum_{k=1}^{N_y/2} \sum_{\ell=1}^{N_z/2} \frac{1}{\left(j^2+k^2+\ell^2\right)^2}.$$

Considering this as a Riemann sum for the integral

$$\int_1^{N_x/2} \int_1^{N_y/2} \int_1^{N_z/2} \frac{1}{(x^2+y^2+z^2)^2}\, dxdydz,$$

which is comparable to

$$\int_1^{\sqrt[3]{N_g}/2} \int_0^{2\pi} \int_0^{\pi} \frac{1}{r^4} r^2 \sin\phi\, d\phi d\theta dr = 2\pi \int_1^{\sqrt[3]{N_g}/2} \frac{1}{r^2}\, dr = 2\pi \left(1 - \frac{2}{\sqrt[3]{N_g}}\right) \approx 2\pi.$$

This finally leads to the result that

$$\sigma_\phi^2 \propto \frac{1}{N_p}.$$

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